

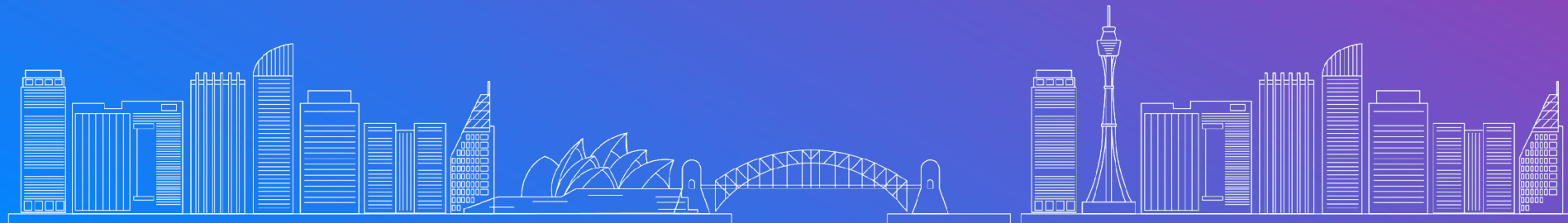
SYDNEY



5 & 6 SEPTEMBER

Build and Scale Kubernetes Platform In Enterprise

Bin Chen



About Me

- 6+ years building Containers Platform (Mesos, Kubernetes)
- 10+ years building Linux & Android System



Build A Basic Kubernetes Cluster is Easy

- Create Kubernetes Control Plane
- Install Add Ons
 - Cluster Autoscaler/Horizontal Pod Autoscaler
 - CNI (Calico, VPC CNI)
 - CSI (EBS, EFS)
 - Ingress Controller
 - Observability(Fluent-Bit/Prometheus/Grafana)
 - Secret
 - Kyverno/Gatekeeper
 - Istio
 - ArgoCD



But, to build a Reliable and Scale Kubernetes Platform, you need to overcome **two hurdles**

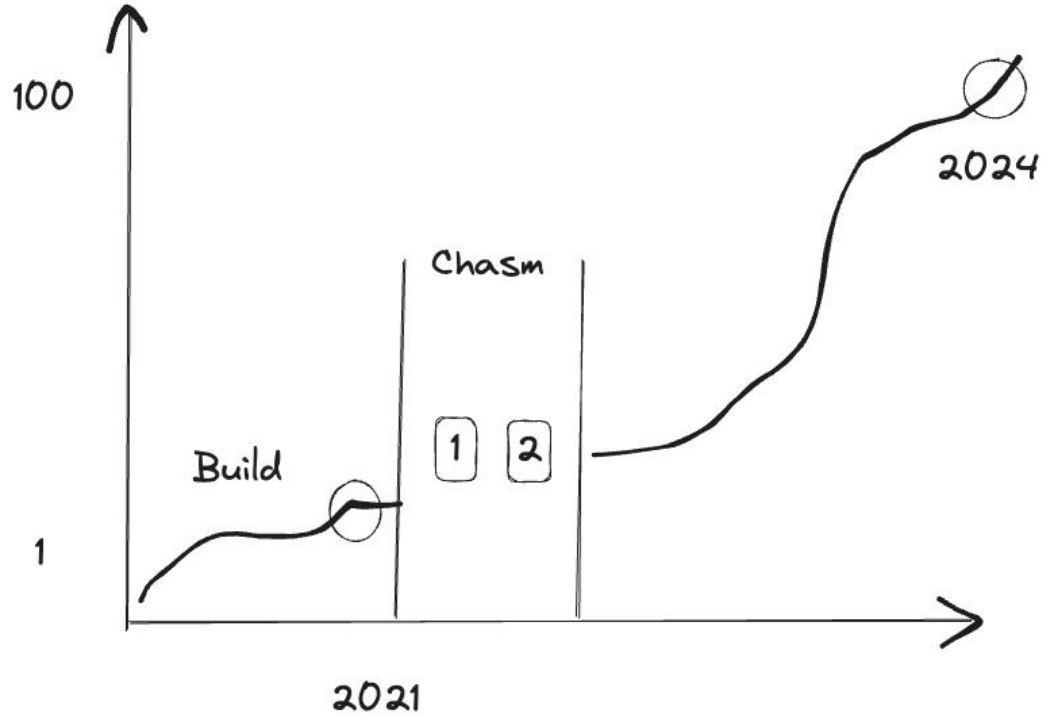
1. Reliability

2. Operation Excellence

- Platform Operation
- Customer Support



Been there, and we crossed it.



What helped?

- Focus
- Think Long Term
- Four Quadrants
 - Feature Development
 - Operating Model
- Fully Committed Team



Focus

Set the boundary, decide what you won't do.

- No CI

Focus on what customer want, do them really well.

- Production clusters, Reliable operation, Be supported

Shield distraction

- Say **No** to things don't align with strategy or priority



Think Long Term

Help you make right/tough decisions for product and team.

Aware broader context: Identify strategic customers, user cases, initiatives

Adapt and Iterate, discovery and address new opportunities



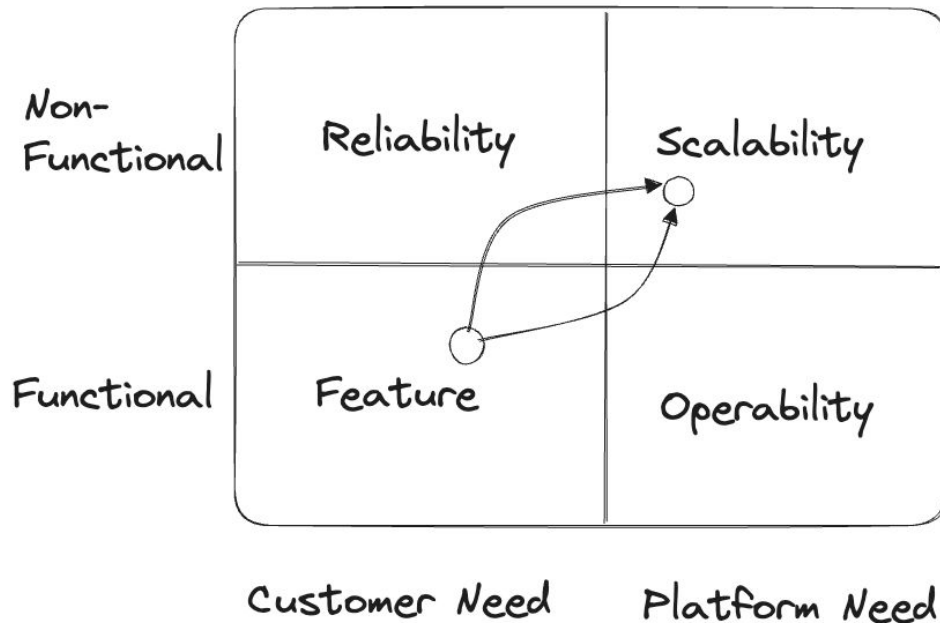
Feature

- 25% time on **features**
- 25% time on **automation**
- Keep **operation excellence** in mind (**build once, operate forever**)
 - Allow only one way for one type of need
 - Limit customer customization to reduce the operation complexity
- Continuous evolvement
 - Karpenter replace Cluster Autoscaler, Kyverno replace Gatekeeper



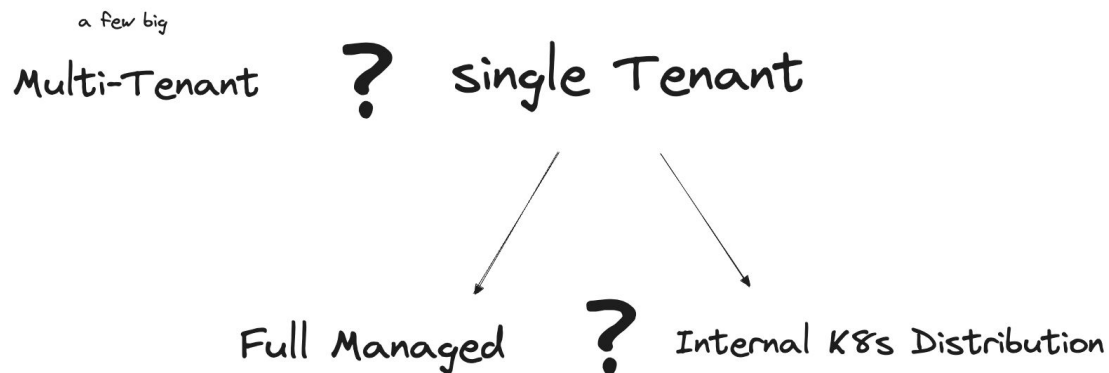
The Four Quadrants(™)

- Two Needs
- Balance
- Define metrics



Platform Operation Excellence

- Operating Model



- Configuration Management

- N cluster * M addons
- GitOps/ArgoCD
- Helm/Helmfile



Fully Committed Team

Ownership

The team is fully empowered to decide **what** to do using disciplined discovery and prioritization approach.

Autonomy with Alignment

Individual can control **how** to do it using established proposal and **decision making process**.



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Thank You!

